KIGALI INDEPENDENT UNIVERSITY (ULK)

Report 1: Project Description Report

Course Assignment: UML Class Diagram Modelling

Project Title: **Student Management System**

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1. Business Idea & Problem Statement

Educational institutions face significant administrative challenges when managing student records, course offerings, and faculty assignments manually or through fragmented systems. These manual processes often result in data discrepancies, delays in enrollment confirmations, and inefficient tracking of academic progress. The proposed **Student Management System (SMS)** is an integrated information system designed to centralize and automate student administration, course distribution, and enrollment tracking to enhance operational efficiency.

2. Purpose of the System

The primary objective of the Student Management System is to provide a unified platform that simplifies administrative workflows. It enables seamless registration of students, systematic organization of academic courses, assignment of qualified instructors, and accurate management of course enrollments.

3. Project Scope

✓ In-Scope (Covered Functions)

- Student profile creation and record management.
- Teacher profile and specialization tracking.
- Course catalog management and credit assignment.
- Processing course enrollments and status updates.

✕ Out-of-Scope (Excluded Functions)

- Tuition fee payments and financial processing.
- Online examination and automated grading.
- Campus housing and hostel accommodation.
- Library book loans and inventory control.

4. Main System Classes & Responsibilities

Student

Maintains personal profiles, academic contact details, and handles student course registration requests.

Teacher

Stores instructor qualifications, specializations, and links faculty to their assigned academic courses.

Course

Defines academic modules, credit values, course codes, and curriculum descriptions.

Enrollment

Acts as the transaction unit linking students to courses, managing status and enrollment dates.